

Markov Chain-Based Authorship Attribution in Koine Greek for Disputed Pauline Letters of the New Testament

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Abstract—This paper examines the authorship of New Testament books traditionally attributed to the Apostle Paul, where scholarly doubts remain. Using texts written in Koine Greek, the analysis models writing style with homogeneous Markov chains in which specific characters represent states. Transition probabilities are estimated from texts of known authorship using maximum likelihood. Each disputed text is then classified according to the author model that yields the highest likelihood. Different modeling approaches and training set sizes are evaluated to identify the most effective method for authorship assessment.

Index Terms—New Testament, Apostle Paul, Koine Greek, authorship attribution, characters, Markov chain, stylometric analysis, probabilistic modeling, text classification.

I. INTRODUCTION

The New Testament constitutes a foundational corpus of early Christian literature, documenting the life and teachings of Jesus Christ and was originally composed in Koine Greek. The Apostle Paul is traditionally attributed authorship of fourteen epistles, accounting for approximately 27% of the New Testament. However, it has been suggested that some letters attributed to Paul may have been written pseudonymously, more precisely, the authorship of seven of these epistles has been questioned due to distinctive stylistic, linguistic and theological characteristics. For instance, the Epistle to the Ephesians demonstrates a stronger emphasis on ecclesiology, adopts a more formal and liturgical style and employs vocabulary not found elsewhere in the New Testament. Similar stylistic and lexical features are observed in the Epistle to the Colossians. Furthermore, the Second Epistle to the Thessalonians contains apparent inconsistencies with statements presented in the First Epistle to the Thessalonians. The Epistle to the Hebrews remains one of the most debated cases of authorship, with multiple proposed authors, including the Apostle Luke and significant stylistic divergence from the traditionally accepted Pauline corpus.

In this paper, we evaluate a character-level Markov model for the classification of Koine Greek texts by authorship. We further examine the impact of removing diacritical marks on

classification accuracy, with the aim of providing insight into the disputed Pauline epistles.

The rest of the paper is organized as follows. Section II discusses related work on the use of Markov models for authorship attribution and on authorship analysis in biblical texts. Section III presents the proposed methodology, including preprocessing, construction of author-specific transition matrices and likelihood-based classification using character-level Markov models. Section IV reports the experimental setup and results, including model selection, accuracy comparisons between diacritical and non-diacritical configurations and the classification of the disputed epistles. Finally, Section V concludes the paper by summarizing the findings and outlining directions for future work.

II. RELATED WORK

Following Markov’s observation in [1] that literary text can be modeled using Markov chains, subsequent studies applied letter- and grammar-based Markov models for authorship identification in English and Russian texts [2], [3].

Leinbach [4] applied a Hidden Markov Model to Koine Greek Deutero-Pauline epistles and found insufficient evidence to reject Pauline authorship. Savoy [5] used Burrows’ Delta, Labbé’s distance and clustering on Ancient Greek and English texts, noting limitations due to small samples. Erickson [6] employed Eder’s Delta and PCA on the Koine Greek New Testament, finding Hebrews closer to Luke–Acts than to Pauline letters. Vocabulary-based measures in [7] suggested clustering among Romans and the Corinthian letters, with Ephesians and Galatians more distant, though not conclusively. Putnikš et al. [8] compared trigram Markov models and function-word MDA on English texts and applied MDA to Hebrews in both Koine Greek and the KJV, concluding it is stylistically inconsistent with Paul and closer to Barnabas in English. Luce and Robertson [9] analyzed the original Koine Greek letters using 20 hand-coded rhetorical features and five similarity metrics, finding strong clustering among the core Pauline letters while Second Thessalonians consistently diverges.

More recent approaches employ machine learning. Using TF-IDF features, [10] finds Hebrews stylistically closer to Luke than to Paul, though near the decision boundary. Similarly, [11] applies TF-IDF and cosine similarity to the Koine Greek New Testament, identifying strong Pauline pairings (Romans–Galatians, Ephesians–Colossians, First Timothy–Titus). A BiLSTM model trained on lemmatized Koine Greek texts in [12] achieves 84% accuracy, classifying Hebrews and First Timothy as non-Pauline, Colossians and Second Thessalonians as Pauline, with mixed results for the remaining epistles.

III. METHODOLOGY

In this section, the modeling of texts similar to [2], will be described, where the set of values of the stochastic process corresponds to the alphabet of the language in which the texts are written (including the space character and other symbols we use to mark end and start of book). For illustrative purposes, the examples provided in this paper are presented in English.

A. Defining State Space

The state space R_X of the Markov chain is defined as

$$R_X = \{l \mid l \text{ is a letter of the English alphabet}\} \cup \{\hat{\cdot}, _, *\},$$

when the analysis is performed on text without diacritical marks. Alternatively,

$$R_X = \{l \mid l \text{ is a letter of the English alphabet}\} \cup \{\hat{\cdot}, _, *\}$$

$$\cup \{d \mid d \text{ is a English letter with a diacritical mark}\},$$

when the analysis is conducted on text that retains diacritical marks. In both cases R_X is a finite set.

In the following subsections we will explain the methodology using simple examples.

B. Preprocessing

Critical editions of the New Testament often contain certain features that may introduce noise in the textual analysis. Therefore, the corpus is preprocessed by removing parentheses and enclosed text, eliminating punctuation and chapter and verse markers, collapsing multiple whitespace characters, converting all characters to lowercase and applying normalization. Optionally, diacritical marks are removed depending on the experimental configuration. Start-of-book and end-of-book symbols are appended, after which tokenization is performed.

Consider the following excerpt: “Love one another.” (John 13:34 NKJV) [13]. Assuming that the text represents a complete book, the preprocessed output is shown below to illustrate the addition of start-of-book and end-of-book symbols:

^love one another*.

The start-of-book and end-of-book symbols are introduced to denote invalid transitions ($* \rightarrow \hat{\cdot}$) when modeling concatenations of multiple books. Additionally, these symbols represent initial and terminal state probabilities within a single book. More specifically, for $n \in \mathbb{N}$, the probability

$$P\{X_n = 1 \mid X_{n-1} = \hat{\cdot}\} = P\{X_n = 1\}$$

represents the probability that the author begins the book with the letter “l.” Likewise, for $n \in \mathbb{N}$, the probability

$$P\{X_n = * \mid X_{n-1} = r\}$$

represents the probability that the author ends the book with the letter “r.” It should be emphasized that the state “*” is absorbing, that is, $p_{*c} = 0$ for all $c \in R_X \setminus \{*\}$, and $p_{**} = 1$.

For the given text, the corresponding sequence of states can be represented as $(X_0, X_1, \dots, X_{16}, X_{17}) = (\hat{\cdot}, l, o, \dots, r, *)$.

C. Construction of Author-Specific Transition Matrices

Let $S \subseteq R_X$ denote the subset of R_X consisting of the characters that appear in the text. For ^love one another*

$$S = \{_, \hat{\cdot}, *, a, e, h, l, n, o, r, t, v\}.$$

As in [2], the transition probabilities are estimated using the maximum likelihood estimator (MLE):

$$\hat{p}_{ij} = \frac{N_{ij}}{\sum_{k \in S} N_{ik}}, \quad (1)$$

where $i, j \in S$, N_{ij} denotes the number of transitions from state i to state j , and $\sum_{k \in S} N_{ik}$ denotes the total number of transitions from state i to any state in S . Furthermore, we assume that all other transition probabilities, for $i \notin S$ or $j \notin S$, are equal to zero.

Estimated non-zero transition probabilities for the example text are presented in Table I.

TABLE I
ESTIMATED NON-ZERO TRANSITION PROBABILITIES FOR THE EXAMPLE TEXT UNDER THE MARKOV MODEL

Transition	p_{ij}	Transition	p_{ij}
$_ \rightarrow a$	0.50	$n \rightarrow e$	0.50
$_ \rightarrow o$	0.50	$n \rightarrow o$	0.50
$\hat{\cdot} \rightarrow l$	1.00	$o \rightarrow n$	0.33
$a \rightarrow n$	1.00	$o \rightarrow t$	0.33
$e \rightarrow _$	0.67	$o \rightarrow v$	0.33
$e \rightarrow r$	0.33	$r \rightarrow *$	1.00
$h \rightarrow e$	1.00	$t \rightarrow h$	1.00
$l \rightarrow o$	1.00	$v \rightarrow e$	1.00

Since the author of the Gospel according to John is traditionally identified as the Apostle John, the transition probabilities above represent a model of his stylistic features.

D. Likelihood Estimation and Text Classification

After constructing stylistic models for multiple authors, we are tasked with classifying a given text according to the author whose style it most closely resembles.

Consider the excerpt: “Love does not envy.” (1 Cor. 13:4 NKJV) [13]. This excerpt has to be classified. The logarithmic probability that the text was authored by a given individual is then computed, as outlined in [2]. Specifically, we evaluate the

log-likelihood that the text was written in the style of John, using the transition probabilities given in Table I:

$$\begin{aligned}\Lambda(t, a = \text{John}) &= \ln P\{o | l\} + \ln P\{v | o\} + \ln P\{e | v\} \\ &\quad + \dots + \ln P\{v | n\} + \ln P\{y | v\} \\ &= \ln(1.00) + \ln(0.33) + \ln(1.00) \\ &\quad + \dots + \ln(0) + \ln(0) \\ &\approx -3.31095.\end{aligned}$$

If regular (non-logarithmic) probabilities were used, multiplying several probabilities would result in extremely small numerical values, making computations impractical. To avoid this issue, we instead compute the sum of logarithmic probabilities. Furthermore, as in [2], transitions that do not appear in the training sets (i.e., those for which $\ln(0)$ would be required) are excluded from the summation.

After computing the corresponding log-likelihood values for each of the remaining authors for whom a model has been constructed, the text is classified as belonging to the author for whom $\Lambda(\text{text}, \text{author})$ attains the maximum value.

In general, the log-likelihood that a text $t = (s_0, s_1, s_2, \dots, s_n)$ was written in the style of author a is computed as

$$\Lambda(t, \text{author} = a) = \sum_{i=0}^{n-1} \ln(p_{s_i, s_{i+1}}), \quad (2)$$

where $s_i, s_{i+1} \in R_X$ for all $i \in \{0, 1, \dots, n-1\}$ and $p_{s_i, s_{i+1}} > 0$.

The classification rule is then defined as

$$C(t) = \arg \max_{a \in A} \Lambda(t, \text{author} = a), \quad (3)$$

where A denotes the set of all authors for whom a Markov chain model has been constructed.

IV. RESULTS

This section presents the dataset, model selection procedure, experimental setup and experimental results.

A. Dataset Overview

In this study, the *Nestle–Aland Novum Testamentum Graece*, 28th edition (NA28) [14], is employed. This edition constitutes a critically reconstructed text based on multiple Koine Greek manuscripts from the relevant historical period and provides a standardized text without orthographic inconsistencies.

B. Model Selection Phase

Initial experiments modeled authors' styles using datasets of varying sizes. The approach was suboptimal, as the model favored the author with the shortest text, leading to fewer transitions and more frequent zero-transition probabilities. Consequently, applying (2) excluded many transitions, treating them as 1 (or 0 in logarithmic terms), preventing the correct adjustment of probabilities.

Using words as states in the Markov chain resulted in consistently low accuracy across different training set sizes.

This was due to a limited dataset, leading to sparse transition matrices and many unseen transitions. On average, 97% of transitions in the test data were absent from the training data, with the extensive Koine vocabulary being a key factor.

Following the preceding discussion, equal-sized training sets are employed and Markov chains with state spaces defined in Section III are considered. In this study, the state space is constructed from the Koine Greek alphabet instead of the illustrative English alphabet.

C. Experimental Setup

We conduct three distinct experiments. In the first, we compare the accuracy of diacritical versus non-diacritical text using undisputed letters as the test set. In the second experiment, the same training sets are utilized to classify the disputed letters from Paul. And in the third, we perform a detailed analysis of the disputed Paul letters that were classified as less decisive in the second experiment. The testing in all three experiments was based on the following principles.

In each test, all authors were assigned the same training set size. Since the authors in the New Testament contribute texts of varying lengths, it was not feasible to include all authors in tests involving larger training sets. Specifically, for larger training sets, classification was performed on a smaller subset of authors. The training set included texts from all other authors as well as the undisputed texts of the Apostle Paul, while his disputed texts were excluded. Additionally, it was assumed that there are no other disputed books in the New Testament aside from the letters of the Apostle Paul.

To ensure that nearly all parts of an author's texts appeared in both the training and test sets and to maintain consistency in testing, 32 classifications were performed for each training set size, with different combinations of training sets. Only in the third experiment we perform 10 classifications instead of 32. The sizes of the test sets varied across classifications. Moreover, the same test cases for a given size were used to test both diacritical and non-diacritical methods.

The books were first preprocessed and subsections were extracted for use as training or test sets.

D. Accuracy Comparison: Diacritical vs. Non-Diacritical Modeling

Testing without diacritical marks is justified by the fact that the original manuscripts did not contain them and that transition probability matrices are expected to be less sparse. Conversely, including diacritical marks is essential to avoid homonyms.

The objective was to identify which modeling approach achieves the highest accuracy and to determine the number of characters for which the model performs best.

A total of ten different training set sizes were tested. To ensure equal representation of each author for a given training set size and to maintain consistency in the number of classifications, the number of tests for each author being classified was approximately uniformly distributed.

As illustrated in Fig. 1, the model incorporating diacritical marks achieves accuracy similar to that of the model without diacritical marks in most cases, with a mean absolute difference in accuracy of 0.07. Although accuracy improves as the training set size increases for both types of models, the model with diacritical marks dominates for larger training sets (with 30000 or more characters), while the model without diacritical marks prevails for smaller training sets. This is due to the increased sparsity of the transition probability matrix when diacritical marks are used with smaller training sets. As the training set size increases, the chances of sparsity in the matrix decrease, allowing the distinction of characters based on diacritical marks to emerge more clearly. In other words, the presence of diacritical marks has greater power to express the author’s style as the training set size increases.

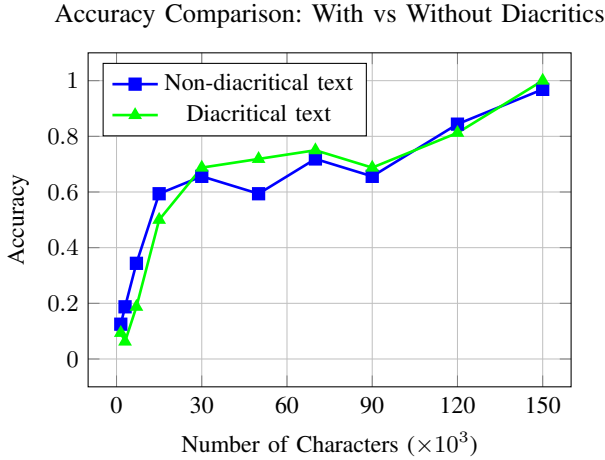


Fig. 1. Accuracy with and without diacritics across different training set sizes

E. Classification of the Disputed Epistles of Saint Paul

To achieve a balance between the number of authors for classification, model accuracy and the involvement of the apostle Paul in the classification process, models are trained using the same training sets of 50000, 70000, 90000 and 120000 characters as described in Section IV-D. These training set sizes are able to evaluate authorship classification among the apostles Matthew, Luke, Mark, John and Paul.

Diagonal values from confusion matrices based on the results presented in Section IV-D for the selected training set sizes are given in Table II. They indicate that the Markov model effectively classifies the texts of the apostle Paul, with rare instances of misclassification. Thus, we anticipate that these tests will yield significant results when applied to the disputed letters of Paul. Additionally, the model provides solid results for the apostles Luke and Matthew. The best performance for the apostle Mark is achieved when using the model with diacritical marks, while results without diacritical marks are notably weaker. Apostle John is consistently identified less frequently and the results for him are significantly less satisfactory. This may be attributed to the fact that his corpus comprises different genres, including narrative and prophecy.

TABLE II
CLASSIFICATION ACCURACY PER AUTHOR FOR NON-DIACRITICAL AND DIACRITICAL TEXT ACROSS SELECTED TRAINING SET SIZES. ENTRIES MARKED AS N/A INDICATE INSUFFICIENT AVAILABLE TRAINING DATA.

Author	Non-diacritical				Diacritical			
	50k	70k	90k	120k	50k	70k	90k	120k
Luke	0.67	0.75	0.88	0.90	0.33	0.75	0.88	1.00
John	0.33	0.38	0.13	0.82	0.50	0.50	0.13	0.64
Paul	1.00	1.00	1.00	0.82	1.00	1.00	1.00	0.82
Matthew	0.50	0.75	0.63	N/A	0.83	0.75	0.75	N/A
Mark	0.43	N/A	N/A	N/A	0.86	N/A	N/A	N/A

When it comes to the disputed letters, each of these letters will be classified using the same training sets that were used for measuring the model’s accuracy. Each letter will be tested in its entirety (no specific sections will be tested separately).

Given that the Apostles Paul and Luke, among all possible candidates, are the only ones to whom the texts were classified, Table III reports the average percentage of times the author of disputed letters is classified as Paul or Luke across all training set sizes. It can be observed that The Second Epistle of Paul to the Thessalonians, The Epistle of Paul to the Colossians and The Epistle of Paul to the Ephesians are consistently classified, by all classifiers in every experimental configuration, as letters authored by the Apostle Paul.

TABLE III
AVERAGE PERCENTAGE OF TIMES THE AUTHOR OF DISPUTED LETTERS IS CLASSIFIED AS PAUL OR LUKE ACROSS ALL TRAINING SET SIZES (32 CLASSIFICATIONS PER CONFIGURATION)

Disputed Letter	Non-Diacritical		Diacritical	
	Luke (%)	Paul (%)	Luke (%)	Paul (%)
1 Timothy	4.69	95.31	0.00	100.00
2 Timothy	7.03	92.97	0.00	100.00
2 Thessalonians	0.00	100.00	0.00	100.00
Colossians	0.00	100.00	0.00	100.00
Ephesians	0.00	100.00	0.00	100.00
Hebrews	69.53	30.47	26.56	73.44
Titus	3.13	96.88	0.00	100.00

In contrast, the remaining four epistles were classified as attributed to Luke in certain instances. Specifically, the First and Second Epistles to Timothy, together with the Epistle to Titus, are classified more frequently as Pauline under all evaluated configurations. The diacritics-based method assigns these three epistles to Paul consistently across all considered training set sizes. Moreover, under the non-diacritics configuration, a small proportion of cases result in attribution to Luke. This distinction is particularly significant, as the diacritics-based method exhibits slightly better classification accuracy relative to the non-diacritics approach for all examined training set sizes, with the sole exception of the 120000-character configuration.

In Fig. 2, we can see that the non-diacritical configuration predominantly classifies the Epistle to the Hebrews, as authored by the Apostle Luke, whereas the diacritical

configuration attributes it to the Apostle Paul in the majority of cases. Notably, in only two experimental configurations, specifically those employing the diacritical setting, the model assigns authorship to Paul with complete consistency, i.e., in all 32 runs. In the remaining configurations, the difference in the number of classifications between the Apostles Paul and Luke is smaller, indicating that the stylistic distinction between the two candidate authors is less clearly defined under those conditions.

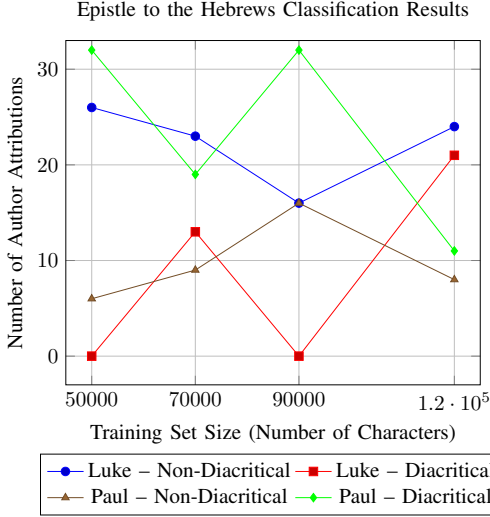


Fig. 2. Classification results for the Epistle to the Hebrews for varying training sizes and preprocessing configurations (32 trials per configuration).

F. Additional Classification Between the Apostles Paul and Luke

In contrast to the previous experiment, which considered multiple authors, the current experiment focuses solely on classifying texts between the Apostles Paul and Luke, as these were the only authors identified in the prior study. This experimental design aims to provide a more detailed analysis of how the selection of specific subsets from each author’s corpus, used as training data, affects the resulting attribution performance. The evaluation consists of 10 test runs per training set size.

As shown in Table IV, the Epistles to the Colossians, Ephesians and the Second Epistle to the Thessalonians were classified as authored by Paul in each test case. The Epistle to Titus and both epistles to Timothy are predominantly classified as Pauline under both methods. Notably, the Epistles to Timothy are classified as Pauline in all cases when diacritics are retained.

An analysis of the training sets shows that, in the majority of cases where the texts were classified as Luke, the training sets for Luke consisted entirely or partially of sections from the book Acts of the Apostles. Approximately 60% of the content of this book focuses on Paul. These passages include Pauline speeches that exhibit rhetorical similarities to both disputed and undisputed epistles. Furthermore, Luke is identified as

TABLE IV
AVERAGE PERCENTAGE OF TIMES THE AUTHOR OF DISPUTED LETTERS IS CLASSIFIED AS PAUL OR LUKE ACROSS ALL TRAINING SET SIZES (10 CLASSIFICATIONS PER CONFIGURATION)

Disputed Letter	Non-Diacritical		Diacritical	
	Luke (%)	Paul (%)	Luke (%)	Paul (%)
1 Timothy	15.00	85.00	0.00	100.00
2 Timothy	20.00	80.00	0.00	100.00
2 Thessalonians	0.00	100.00	0.00	100.00
Colossians	0.00	100.00	0.00	100.00
Ephesians	0.00	100.00	0.00	100.00
Hebrews	70.00	30.00	22.50	77.50
Titus	22.50	77.50	7.50	92.50

Paul’s companion in Colossians 4:14 (NKJV) [13] and 2 Timothy 4:11 (NKJV) [13]. These factors may explain why the likelihood of Luke being the author is highest in certain cases and why his likelihood ranked second in the results presented in Section IV-E.

Concerning the Epistle to the Hebrews, non-diacritical method favors Luke, while diacritical method favors Paul, as shown in Fig. 3, with the latter appearing more decisive, favoring Paul in 77.5% of cases. Moreover, in approximately 21% of the test cases where the disputed letter was attributed to Luke, Luke’s training set consisted of sections from the Acts of the Apostles, which may suggest that Luke could indeed be the author. However, further analysis is needed to draw more definitive conclusions.

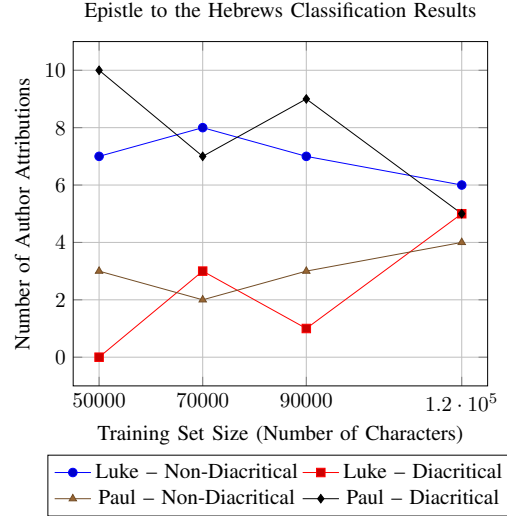


Fig. 3. Classification results for the Epistle to the Hebrews for varying training sizes and preprocessing configurations (10 trials per configuration).

V. CONCLUSION AND FUTURE WORK

The results demonstrate that character-level Markov modeling is a viable and effective method for authorship attribution in Koine Greek texts, especially when larger training datasets are used. The inclusion or exclusion of diacritical marks does

not significantly affect overall performance. Author identification accuracy varied across authors. The Apostle Paul was identified with the highest accuracy, followed by Luke and Matthew, while Mark and especially John showed lower identification rates.

Several disputed epistles, Second Thessalonians, Colossians and Ephesians, were consistently classified as Pauline across all test cases. The Epistles to Timothy and Titus were attributed to Paul in the majority of experiments. Misclassifications, particularly those involving Luke as a secondary match, may be influenced by the presence of Pauline speeches and narratives within Luke's training corpus. Further investigation is required to reach a more definitive conclusion regarding the Epistle to the Hebrews.

At the same time, the conclusions should be interpreted with caution. The absence of smoothing for unseen transitions, the restriction to a small set of candidate authors, and the influence of shared textual content (e.g., Luke's association with Paul) may affect the classification outcomes. Overall, the findings should be viewed as indicative rather than definitive.

Future work may include examining the effect of smoothing transition probabilities, analyzing the impact of assigning a "penalty" probability to transitions that do not occur, comparing results using additional evaluation metrics, conducting a baseline comparison with simpler classification methods under identical experimental conditions, and investigating the influence of certain words from the texts. It would also be beneficial to explore in greater detail the reasons why the model rarely identifies the Apostle John as the author of his texts. Another possible direction is to incorporate authors whose writings are not part of the New Testament but are mentioned within it (e.g., Barnabas). Finally, a more detailed analysis of the authorship of the Epistle to the Hebrews could be performed.

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